MUTHU VARSHITH

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PROFESSIONAL SUMMARY

AI/ML Engineer with a strong Python foundation, building LLM agents, retrieval augmented generation systems and deep learning models delivered as working services. Fine tuned and evaluated vision and speech models in PyTorch, built LangGraph agents that keep a human in the loop, and measured design choices on labelled benchmarks with significance testing. Ships with FastAPI, PostgreSQL, Docker and CI, backed by more than 1,000 automated tests across three projects.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, Object Oriented Programming
Machine Learning: Scikit Learn, Supervised Learning, Classification, Model Evaluation, AUC, Pandas, NumPy
Deep Learning: PyTorch, Hugging Face Transformers, Computer Vision, NLP, LoRA Fine Tuning
Generative AI: LLM Agents, Tool Calling, LangGraph, LangChain, RAG, Hybrid Retrieval, Embeddings, Rerankers, Prompt Engineering, RAGAS, LLM as a Judge
Backend and Databases: FastAPI, Pydantic, REST APIs, PostgreSQL, pgvector, Qdrant, SQLite
Deployment and Tools: MLOps, MLflow, ONNX, Model Quantisation, Docker, GitHub Actions, AWS, Terraform, Langfuse, Git, pytest

PROJECTS

Cerberus | **Agentic Software Repair with a Verification Gate** *Aug 2026 – Present*

- Built an agentic bug repair system where a tool calling LLM agent proposes patches and a deterministic gate decides, running RED, GREEN, regression and scope checks in an offline Docker sandbox; 10 of 10 valid repairs were admitted with zero false admissions.
- Raised bug localization Recall@10 from 0.49 to 0.86 on 120 SWE Bench Verified instances using Tree sitter chunking, BM25, Qdrant and Reciprocal Rank Fusion.
- Shipped a FastAPI GitHub App with LangGraph orchestration and human approval before any pull request; the gate cut wrong approvals from 65% to 36%.

HireAI | **AI Recruiting Platform with RAG and Human Approval** *Jul 2026 – Present*

- Built a FastAPI platform on PostgreSQL and pgvector with a LangGraph agent whose shortlists are saved only after human approval, and RAG answers that cite resume passages.
- Measured hybrid retrieval with vector search, BM25, Reciprocal Rank Fusion and cross encoder reranking on a 30 question labelled benchmark, keeping vector search at MRR@10 of 0.833.
- Secured it with argon2id hashing, hashed sessions, rate limited login and tested prompt injection guards; delivered Docker Compose and GitHub Actions CI with 639 passing tests.

FusionTrace | **Privacy First Deepfake Detection Service** *Mar 2025 – Present*

- Built a FastAPI and PyTorch service that scores images, audio and video on local hardware, fusing swappable detectors into one weighted, explained risk score.
- Evaluated models on unseen external data; a ViT image detector reached 98.9% accuracy on AI generated faces and a Wav2Vec2 audio model 98.3% on a 4,002 clip held out set.
- Added LoRA fine tuning with leakage aware data splits, MLflow experiment tracking and model registry, ONNX export and INT8 quantisation.

WORK EXPERIENCE

Python Full Stack Development Intern | **KodNest** *Feb 2026 – May 2026*

- Developed a Python inventory, sales and purchase management system covering stock management, multi item invoicing, returns and payments, with role based access control for admin, sales and inventory teams.
- Implemented stock and billing logic in atomic database transactions so inventory never goes negative, and validated the business rules with 59 automated tests covering overselling, returns and access control.

Artificial Intelligence Intern | **SmartEd Solutions** *Jun 2025 – Oct 2025*

- Trained and compared Logistic Regression and SVM classifiers in Scikit Learn on the 569 sample Breast Cancer Wisconsin Diagnostic dataset after exploratory data analysis and feature standardization.
- Achieved 97.4% held out accuracy with Logistic Regression versus 95.6% with SVM on an 80/20 split, evaluated with confusion matrices and ROC curves.

EDUCATION

Visvesvaraya Technological University | B.E. in Computer Science and Business Systems *2022 – 2026*

CERTIFICATIONS

Oracle 2025: OCI Generative AI Professional · OCI Data Science Professional · Fusion AI Agent Studio Associate
AWS: Foundations of Prompt Engineering · **Nasscom Foundation:** Machine Learning & Image Processing